

Runsang Liu

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Education

Ph.D. May 2027 (Expected)	Dual-Title in Industrial Engineering & Operations Research The Pennsylvania State University Advisor: Dr. Hui Yang
M.S. May 2020	Dual-Title in Industrial Engineering & Operations Research Minor in Computational Science The Pennsylvania State University
B.S. May 2018	Systems Engineering University of New Haven

Research Interests

Physical, statistical, and data-driven modeling of cyber-physical systems for process monitoring and control, digital twins, process improvement, and intelligent automation, with applications in advanced manufacturing and healthcare. My specific research interests include:

- Sensor-based monitoring, modeling, and control of advanced manufacturing processes
- Physics-based modeling for efficient simulation of multiphysics dynamics
- Multiscale statistical modeling for process understanding and improvement
- Digital twin modeling, synchronization, and operational support for cyber-physical systems
- Data-driven analytics and AI-enabled manufacturing decision support

Honors & Awards

- **2026, Finalist**, IISE M&D Best Student Paper Award, Arlington, TX.
- **2026, Participant, Doctoral Student Colloquium**, 2026 IISE Annual Conference, Arlington, TX.
- **2025, Marcus Scholarship for outstanding doctoral students**, Department of Industrial and Manufacturing Engineering, The Pennsylvania State University.
- **2025, Cover Article**, *Chaos: An Interdisciplinary Journal of Nonlinear Science*, Volume 35, Issue 4.
- **2023, Finalist**, IISE M&D Best Student Paper Award, New Orleans, LA.
- **2021, 3rd Place**, ASME Computers and Information in Engineering Student Hackathon.
- **2020, NSF INTERN Scholarship**, IBM Thomas J. Watson Research Center.
- **2016, Outstanding Senior in Systems Engineering**, University of New Haven.

Publications

Journal Articles

1. **R. Liu**, A. Iquebal, and H. Yang (2026). “Optimizing Self-organizing Network for Thermal Simulation in Additive Manufacturing.” *Journal of Manufacturing Science and Engineering*, 148(10), p. 101010.[doi:10.1115/1.4072313](https://doi.org/10.1115/1.4072313) (Best Student Paper Award Finalist in the 2026 IISE Annual Conference M&D Division)
2. **R. Liu** and H. Yang (2026). “Wavelet Gaussian Process for Multi-scale Monitoring of Imaging Data.” *IISE Transactions*, pp. 1–14. [doi:10.1080/24725854.2026.2632144](https://doi.org/10.1080/24725854.2026.2632144)
3. X. Peng, **R. Liu**, and H. Yang (2026). “STEM: STructural EMbedding for Graph Representation of 3D Designs.” *Manufacturing Letters*. Accepted.
4. **R. Liu** and H. Yang (2025). “Self-organizing Network Simulation of Cardiac Contraction Dynamics.” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(12), p. 121105. [doi:10.1063/5.0305451](https://doi.org/10.1063/5.0305451)
5. **R. Liu** and H. Yang (2025). “Self-organizing Network Simulation of Cardiac Electrical Dynamics.” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(4), p. 041102. [doi:10.1063/5.0261019](https://doi.org/10.1063/5.0261019) (Cover Article)
6. **R. Liu** and H. Yang (2024). “Self-organizing Network Representation of Human Heart.” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(12), pp. 1–7. [doi:10.1063/5.0243391](https://doi.org/10.1063/5.0243391)
7. **R. Liu** and H. Yang (2023). “Multimodal Probabilistic Modeling of Melt Pool Geometry Variations in Additive Manufacturing.” *Additive Manufacturing*, 61, p. 103375. [doi:10.1016/j.addma.2022.103375](https://doi.org/10.1016/j.addma.2022.103375) (Best Student Paper Award Finalist in the 2023 IISE Annual Conference M&D Division)
8. **R. Liu**, B. D. Vogt, and H. Yang (2021). “Gaussian Process Monitoring of Layerwise-dependent Imaging Data.” *IEEE Robotics and Automation Letters*, 6(4), pp. 8029–8036. [doi:10.1109/LRA.2021.3102625](https://doi.org/10.1109/LRA.2021.3102625)
9. **R. Liu** and H. Yang (2021). “Spread Tessellation of Infectious Diseases for Epidemic Decision Support.” *IEEE Robotics and Automation Letters*, 7(1), pp. 626–633. [doi:10.1109/LRA.2021.3131699](https://doi.org/10.1109/LRA.2021.3131699)
10. H. Yang, **R. Liu**, and S. Kumara (2020). “Self-organizing Network Modelling of 3D Objects.” *CIRP Annals*, 69(1), pp. 409–412. [doi:10.1016/j.cirp.2020.04.099](https://doi.org/10.1016/j.cirp.2020.04.099)

Manuscripts Under Review

1. **R. Liu** and H. Yang (2026). “Self-organizing Network for Personalized Modeling of Heart Geometry.” *Chaos: An Interdisciplinary Journal of Nonlinear Science*.
2. **R. Liu** and H. Yang (2026). “Digital Twinning and Calibration of Cardiac Electrical Dynamics.” *INFORMS Journal on Data Science*.

Peer-reviewed Conference Papers

1. **R. Liu** and H. Yang (2026). “Self-organizing Network Simulation of Multiphysics Dynamics: Framework, Formulation, and Applications.” *2026 Winter Simulation Conference*. Accepted.
2. **R. Liu** and H. Yang (2025). “Optimizing Network Simulation of Cardiac Electrical Dynamics.” In *Proceedings of the 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, August 17–21, Los Angeles, CA, pp. 814–819. doi:10.1109/CASE58245.2025.11163831
3. X. Peng, **R. Liu**, and H. Yang (2025). “SCOUT: Spatiotemporal Coverage for Optimal Unmanned Tasking.” In *Proceedings of the 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, August 17–21, Los Angeles, CA, pp. 2103–2108. doi:10.1109/CASE58245.2025.11163795
4. H. Yang, S. Zhang, **R. Liu**, A. Drall, Y. Wang, M. Ventura, and C. Deflitch (2021). “Epidemic Informatics and Control: A Holistic Approach from System Informatics to Epidemic Response and Risk Management in Public Health.” In *AI and Analytics for Public Health—Proceedings of the 2020 INFORMS International Conference on Service Science*. Springer, Chapter 1, pp. 1–46. doi:10.1007/978-3-030-75166-1_1

Conference Talks & Presentations

1. **R. Liu**, A. Iquebal, and H. Yang, “Optimizing Self-organizing Network for Thermal Simulation in Additive Manufacturing,” *IISE Annual Conference 2026*, May 16–19, 2026, Arlington, TX.
2. **R. Liu** and H. Yang, “Wavelet Gaussian Process for Multi-scale Monitoring of Imaging Data,” *2025 INFORMS Annual Meeting*, October 26–29, 2025, Atlanta, GA.
3. **R. Liu** and H. Yang, “Optimizing Network Simulation of Cardiac Electrical Dynamics,” *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, August 16–21, 2025, Los Angeles, CA.
4. X. Peng, **R. Liu**, and H. Yang, “Spatiotemporal Coverage for Optimal Unmanned Tasking,” *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, August 16–21, 2025, Los Angeles, CA.
5. H. Yang, T. Kuo, and **R. Liu**, “Network Science for Cyber-physical Twinning of Human Heart,” *NIH-sponsored Special Track on Cyberinfrastructure and Emerging Technologies in Biomedical Research at ISMB/ECCB 2025*, Liverpool, United Kingdom. (Presenter: Dr. H. Yang.)
6. **R. Liu** and H. Yang, “Wavelet Gaussian Process for Multi-scale Monitoring of Imaging Data” (poster), *IISE Annual Conference 2025*, May 31–June 3, 2025, Atlanta, GA.
7. **R. Liu** and H. Yang, “Self-organizing Network Simulation of Cardiac Electrical Dynamics,” *Penn State Graduate Exhibition 2025*, March 28, 2025, The Pennsylvania State University.
8. **R. Liu** and H. Yang, “Recurrence Network Modeling for Cardiac Digital Twin,” *2024 INFORMS Annual Meeting*, October 18–21, 2024, Seattle, WA.

9. **R. Liu** and H. Yang, “Multi-modal Probabilistic Modeling and Characterization of Melt Pools,” *2023 INFORMS Annual Meeting*, October 15–18, 2023, Phoenix, AZ.
10. **R. Liu** and H. Yang, “Spatial Tessellation of Infectious Disease Spread for Epidemic Decision Support,” *IISE Annual Conference 2023*, May 21–24, 2023, New Orleans, LA.
11. **R. Liu** and H. Yang, “Multimodal Probabilistic Modeling of Melt Pool Geometry Variations in Additive Manufacturing,” *IISE Annual Conference 2023*, May 21–24, 2023, New Orleans, LA.
12. **R. Liu** and H. Yang, “Photon Distribution Estimation for Melt Pool Characterization,” *Additive Manufacturing & Design (AMD) Poster Session*, 2022, Penn State University.
13. **R. Liu**, B. Vogt, and H. Yang, “Spatiotemporal Gaussian Process for Layerwise Monitoring of Additive Manufacturing,” *17th IEEE International Conference on Automation Science and Engineering (CASE)*, August 23–27, 2021, Lyon, France.
14. **R. Liu** and H. Yang, “Spatiotemporal Gaussian Process for Layerwise Monitoring of Additive Manufacturing,” *2021 INFORMS Annual Meeting*, November 13, 2021, Virtual.
15. H. Yang, **R. Liu**, and S. Kumara, “Self-organizing Network Modelling of 3D Objects,” *ICDS Symposium Poster Session*, 2020, Penn State University.

Teaching Experience

Instructor, Industrial and Manufacturing Engineering

- IE 424: Process Quality Engineering (Spring 2022)
 - Taught a 3-credit undergraduate engineering course with 116 students enrolled.
 - Assumed full responsibility for the course, including designing course materials, preparing and delivering lectures.
 - Created homework assignments, quizzes, and examinations.

Teaching Assistant, Industrial and Manufacturing Engineering

- IE 453: Simulation Modeling for Decision Support (Fall 2022)
 - Led two hands-on Simio laboratory sessions for 53 students.
- IE 516: Applied Stochastic Processes (Spring 2020; Spring 2021)

Research Experience

Self-organizing Networks for Multiphysics Modeling of Complex Systems

Research Assistant (*Sponsored by NIH R01 HL172292 and NSF CMMI-2531899*)

- Cardiac Multiphysics Modeling
 - Designed a novel self-organizing network representation of cardiac geometry.
 - Developed computational methodologies for simulating spatiotemporal electrical dynamics and mechanical contraction on the network representation.
 - Established a mesh-free and compact framework for representing cardiac geometry and multiphysics dynamics in cyberspace, supporting personalized cardiac digital twins and

targeted treatment planning.

- Thermal Modeling in Additive Manufacturing
 - Developed a self-organizing network framework for modeling heat accumulation and transfer in geometrically complex parts.
 - Designed computational methodologies for model calibration.
 - Established a generalized framework for optimal design and calibration of network-based thermal models, demonstrating strong potential for improving the prediction and control of advanced manufacturing processes.
- **Research Outputs & Recognition**
 - Four peer-reviewed journal publications in leading journals
 - *Journal of Manufacturing Science and Engineering*
 - *Chaos: An Interdisciplinary Journal of Nonlinear Science*
 - One peer-reviewed conference publication.
 - Cover article in *Chaos: An Interdisciplinary Journal of Nonlinear Science*.
 - Finalist in the 2026 IISE Manufacturing & Design Best Student Paper Competition.

Sensor-based Modeling, Monitoring, and Control of Additive Manufacturing
Research Assistant (*Sponsored by NSF CMMI-2531899, NSF CMMI-2011289, and NIST Cooperative Agreement 70NANB19H127*)

- In Situ Layerwise Inspection and Quality Monitoring in Additive Manufacturing
 - Develop statistical modeling and monitoring methodologies for in situ quality assessment of additive manufacturing processes using layerwise imaging data.
 - Develop a wavelet-based statistical framework for multiscale analysis of variance (ANOVA) and image-guided statistical process control (SPC).
 - Establish a data-driven framework for monitoring layer-to-layer process evolution and identifying abnormal process behavior during additive manufacturing.
- Probabilistic Modeling and Characterization of Melt Pool Geometry
 - Develop a novel approach to simulate photon emission for statistical estimation and modeling of the multimodal probability distribution of a melt pool.
 - Develop a multimodal probabilistic model to characterize the complex geometric distribution and variability of melt pools from in situ imaging data.
 - Quantify the effects of process conditions on melt pool geometry and variability, providing insights into process-geometry relationships.
- **Research Outputs & Recognition**
 - Three peer-reviewed journal publications in leading journals
 - *IISE Transactions*
 - *Additive Manufacturing*
 - *IEEE Robotics and Automation Letters*
 - Finalist in the 2023 IISE Manufacturing & Design Best Student Paper Competition.

Invited Guest Lecture

- **R. Liu** (2026). “Multimodal Probabilistic Modeling of Melt Pool Geometry Variations in Additive Manufacturing.” IE 484: Introduction to Maintainability Engineering, **University of Tennessee, Knoxville**.

Related News

- March 2026, “Engineering a Better Heartbeat,” **Penn State News**.
- September 2020, “Stitching It Together: Two Engineering Students Spend Summer with IBM,” **Penn State News**.
- May 2020, “Industrial Engineering Department Pivots to Address COVID-19,” **Penn State News**.

Mentoring

Undergraduate Students

(Funded by NSF REU Program)

- Alex Luna-Ow (Summer 2025), “Network Simulation of Cardiomyocyte Contraction Dynamics”. The abstract was accepted for poster presentation at the 2025 BMES Annual Meeting in San Diego, CA.
- Ho Chung Cheung (Fall 2019-Spring 2020), “Virtual Reality for Smart Manufacturing”.
- Davina Stamp (Spring 2020), “Simulation of Action Potential of a Myocyte in Virtual Reality”.

Graduate Student

- Xinghao Peng (Fall 2024-present)
 - “SCOUT: Spatiotemporal Coverage for Optimal Unmanned Tasking”, published in *Proceedings of the 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*.
 - “STEM: STructural EMbedding for Graph Representation of 3D Designs”, accepted by *Manufacturing Letters*.

Service Activities

- 2025–2026, **Vice President**, ASQ Student Chapter, The Pennsylvania State University.
- 2024–2025, **Training Assistant**, Asian Student Fitness Club, The Pennsylvania State University.
- 2024, **Session Chair**, “Data Assimilation for Simulation Optimization,” 2024 INFORMS Annual Meeting.

Referee

- *ASME Journal of Manufacturing Science and Engineering*

- *IEEE Robotics and Automation Letters*
- *IEEE Transactions on Biomedical Engineering*
- *IEEE Journal of Biomedical and Health Informatics*
- *IEEE Conference on Automation Science and Engineering*

Professional Membership

- Institute of Industrial and Systems Engineers (IISE)
 - Quality Control and Reliability Engineering (QCRE)
 - Data Analytics and Information Systems (DAIS)
 - Manufacturing and Design (M&D)
- Institute for Operations Research and the Management Sciences (INFORMS)
 - Quality, Statistics, and Reliability Section
 - Data Mining Society
 - Simulation Society
- Institute of Electrical and Electronics Engineers (IEEE)
 - Robotics and Automation Society